

Airflow - Introduction and installation

Installation:

Fetch provided docker-compose.yml from DataScientest S3 bucket

```
ubuntu@ip-172-31-46-112:~/datascientest-data-engineering/airflow$ wget https://dst-de.s3.eu-west-3.amazonaws.com/airflow_fr/docker-compose/docker-compose.yml
--2025-07-03 17:12:54-- https://dst-de.s3.eu-west-3.amazonaws.com/airflow_fr/docker-compose/docker-compose.yml
Resolving dst-de.s3.eu-west-3.amazonaws.com (dst-de.s3.eu-west-3.amazonaws.com)... 3.5.205.182, 52.95.155.64
Connecting to dst-de.s3.eu-west-3.amazonaws.com (dst-de.s3.eu-west-3.amazonaws.com)|3.5.205.182|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 5277 (5.2K) [binary/octet-stream]
Saving to: 'docker-compose.yml'

docker-compose.yml 100%[=====] 5.15K --.-KB/s in 0s

2025-07-03 17:12:54 (108 MB/s) - 'docker-compose.yml' saved [5277/5277]
```

Create volume directories and set the permissions. Airflow stores DAGs, logs, and plugins by bind-mounting host directories into containers.

```
ubuntu@ip-172-31-46-112:~/datascientest-data-engineering/airflow$ mkdir ./dags ./logs ./plugins
ubuntu@ip-172-31-46-112:~/datascientest-data-engineering/airflow$ sudo chmod -R 777 dags/ logs/ plugins/
```

Generate the .env file

```
ubuntu@ip-172-31-46-112:~/datascientest-data-engineering/airflow$ echo -e "AIRFLOW_UID=$(id -u)\nAIRFLOW_GID=0" > .env
ubuntu@ip-172-31-46-112:~/datascientest-data-engineering/airflow$ cat .env
AIRFLOW_UID=1000
AIRFLOW_GID=0
```

initialise the database to use Airflow

```
ubuntu@ip-172-31-46-112:~/datascientest-data-engineering/airflow$ docker-compose up airflow-init
Building with native build. Learn about native build in Compose here: https://docs.docker.com/go/compose-native-build/
Creating network "airflow_default" with the default driver
Creating volume "airflow_postgres-db-volume" with default driver
Pulling postgres (postgres:13)...
13: Pulling from library/postgres
5695a9e05cd: Pull complete
194596813730: Pull complete
226e2e2e0225: Pull complete
c43e529dbdb4: Pull complete
2b98c45d8b02: Pull complete
99298412a2b8: Pull complete
e3175991a815: Pull complete
896a888932e0: Pull complete
af6894885a5e: Pull complete
84f963838a13: Pull complete
67c5a8a5c3b0: Pull complete
f8786a3bfaff: Pull complete
ca07d2d388e6: Pull complete
ba021985985: Pull complete
Digest: sha256:1b5a2a75dec38df4835184989451b89e5a2e87d98e69e1c905ace8b941bbdbf4
Status: Downloaded newer image for postgres:13
Pulling redis (redis:latest)...
latest: Pulling from library/redis
5695a9e05cd: Already exists
d6655ba2ccca: Pull complete
4ef8ef7693bb: Pull complete
881b4a6f26cc: Pull complete
667393f5b310: Pull complete
4440708e154: Pull complete
ab22bb3686ca: Pull complete
Digest: sha256:b42dc0db19e152e2abf37a53cf7983522cfff056a258589697a9b6c11
Status: Downloaded newer image for redis:latest
Pulling airflow-init (apache/airflow:2.8.1)...
2.8.1: Pulling from apache/airflow
e1cac4e69b2: Pull complete
51d1f07986b7: Pull complete
979549a88b2: Pull complete
f86bc27bf61: Pull complete
1a586c2dc01: Pull complete
e252f191f43: Pull complete
36e2e2638b2: Pull complete
1e4eebfab5d: Pull complete
eda294da76c9: Pull complete
e27351ba3c0f: Pull complete
a295dc39a81: Pull complete
72a9d6a8ab48: Pull complete
6a38a7e7e19: Pull complete
c6b311e165e8: Pull complete
d2517f65443: Pull complete
4716dc5de2: Pull complete
ae71fc46f15b: Pull complete
2876a994157: Pull complete
8023278e993c: Pull complete
7145897c134d: Pull complete
1cc0f68fe59a: Pull complete
4747b78ef54: Pull complete
Digest: sha256:c636ad08b8ee987c4b4b363b3b726451984d298dcab67bd2957cd533bac
Status: Downloaded newer image for apache/airflow:2.8.1

Creating airflow_postgres 1 ... done
Creating airflow_redis 1 ... done
Creating airflow-init 1 ... done
Attaching to airflow_redis-1, airflow-init-1
airflow-init-1 | /home/airflow/.local/lib/python3.8/site-packages/airflow/configuration.py:812 DeprecationWarning: The sqlalchemy_conn option in [core] has been moved to the sqlalchemy_conn option in [database] - the old setti
airflow-init-1 | ng has been used, but please update your config.
airflow-init-1 | /home/airflow/.local/lib/python3.8/site-packages/airflow/configuration.py:724 DeprecationWarning: The sqlalchemy_conn option in [core] has been moved to the sqlalchemy_conn option in [database] - the old setti
airflow-init-1 | ng has been used, but please update your config.
airflow-init-1 | /home/airflow/.local/lib/python3.8/site-packages/airflow/configuration.py:747 DeprecationWarning: The auth_backend option in [api] has been renamed to auth_backends - the old setting has been used, but please up
airflow-init-1 | date your config.
airflow-init-1 | /home/airflow/.local/lib/python3.8/site-packages/airflow/configuration.py:763 FutureWarning: The auth_backend setting in [api] has had airflow.api.auth_backend.session added in the running config, which is need
airflow-init-1 | ed by the UI. Please update your config before Apache Airflow 3.0.
airflow-init-1 | /home/airflow/.local/lib/python3.8/site-packages/airflow/configuration.py:738 DeprecationWarning: The sqlalchemy_conn option in [core] has been moved to the sqlalchemy_conn option in [database] - the old setti
airflow-init-1 | ng has been used, but please update your config.
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airflow-init-1 | date your config.
airflow-init-1 | /home/airflow/.local/lib/python3.8/site-packages/airflow/models/base.py:71 DeprecationWarning: The sqlalchemy_conn option in [core] has been moved to the sqlalchemy_conn option in [database] - the old setti
airflow-init-1 | ng has been used, but please update your config.
airflow-init-1 | DB: postgresql+psycopg2://airflow:***@postgres/airflow
airflow-init-1 | Performing upgrade to the metadata database postgresql+psycopg2://airflow:***@postgres/airflow
airflow-init-1 | [2025-07-03T17:20:28.854+0000] [migration.py:210] INFO - Context impl PostgresqlImpl.
airflow-init-1 | [2025-07-03T17:20:28.854+0000] [migration.py:210] INFO - Will assume transactional DDL.
airflow-init-1 | [2025-07-03T17:20:28.854+0000] [migration.py:210] INFO - Context impl PostgresqlImpl.
airflow-init-1 | [2025-07-03T17:20:28.854+0000] [migration.py:210] INFO - Will assume transactional DDL.
airflow-init-1 | INFO [alembic.runtime.migration] Context impl PostgresqlImpl.
airflow-init-1 | INFO [alembic.runtime.migration] Will assume transactional DDL.
airflow-init-1 | INFO [alembic.runtime.migration] Running stamp_revision -> 83344c1d9334
airflow-init-1 | Database migrating done!
airflow-init-1 | WARNING: Environment variable 'AIRFLOW_DB_UPGRADE' is deprecated please use 'AIRFLOW_DB_MIGR
airflow-init-1 | ATE' instead
airflow-init-1 | /home/airflow/.local/lib/python3.8/site-packages/airflow/configuration.py:812 DeprecationWarning: The sqlalchemy_conn option in [core] has been moved to the sqlalchemy_conn option in [database] - the old setti
airflow-init-1 | ng has been used, but please update your config.
airflow-init-1 | /home/airflow/.local/lib/python3.8/site-packages/airflow/configuration.py:724 DeprecationWarning: The sqlalchemy_conn option in [core] has been moved to the sqlalchemy_conn option in [database] - the old setti
airflow-init-1 | ng has been used, but please update your config.
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airflow-init-1 | date your config.
airflow-init-1 | /home/airflow/.local/lib/python3.8/site-packages/airflow/configuration.py:763 FutureWarning: The auth_backend setting in [api] has had airflow.api.auth_backend.session added in the running config, which is need
airflow-init-1 | ed by the UI. Please update your config before Apache Airflow 3.0.
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airflow-init-1 | date your config.
airflow-init-1 | /home/airflow/.local/lib/python3.8/site-packages/airflow/models/base.py:71 DeprecationWarning: The sqlalchemy_conn option in [core] has been moved to the sqlalchemy_conn option in [database] - the old setti
airflow-init-1 | ng has been used, but please update your config.
airflow-init-1 | [2025-07-03T17:20:28.854+0000] [override.py:1369] INFO - Inserted Role: Admin
airflow-init-1 | [2025-07-03T17:20:28.854+0000] [override.py:1369] INFO - Inserted Role: Public
airflow-init-1 | [2025-07-03T17:20:28.854+0000] [override.py:868] WARNING - No user yet created, use flask fab
airflow-init-1 | command to do it.
```

```

airflow-init_1 | [2025-07-03T17:20:28.147+0000] [override.py:1769] INFO - Created Permission View: can edit on
airflow-init_1 | Passwords
airflow-init_1 | s to role Admin
airflow-init_1 | Passwords
airflow-init_1 | s to role Admin
airflow-init_1 | My Password
airflow-init_1 | ord to role Admin
airflow-init_1 | My Password
airflow-init_1 | ord to role Admin
airflow-init_1 | My Profile
airflow-init_1 | le to role Admin
airflow-init_1 | My Profile
airflow-init_1 | le to role Admin
airflow-init_1 | n Users
airflow-init_1 | to role Admin
airflow-init_1 | Users
airflow-init_1 | role Admin
airflow-init_1 | Users
airflow-init_1 | to role Admin
airflow-init_1 | Users
airflow-init_1 | on Security
airflow-init_1 | ity to role Admin
airflow-init_1 | n Roles
airflow-init_1 | to role Admin
airflow-init_1 | Roles
airflow-init_1 | role Admin
airflow-init_1 | n Roles
airflow-init_1 | to role Admin
airflow-init_1 | on List Roles
airflow-init_1 | Roles to role Admin

```

```

airflow-init_1 | [2025-07-03T17:20:28.004+0000] [override.py:1769] INFO - Created Permission View: can read on
airflow-init_1 | User Stats Chart
airflow-init_1 | s to role Admin
airflow-init_1 | User's Statistics
airflow-init_1 | s to role Admin
airflow-init_1 | Permissions
airflow-init_1 | ons to role Admin
airflow-init_1 | on Actions
airflow-init_1 | os to role Admin
airflow-init_1 | View Menus
airflow-init_1 | us to role Admin
airflow-init_1 | on Resources
airflow-init_1 | res to role Admin
airflow-init_1 | Permission Views
airflow-init_1 | on Views to role Admin
airflow-init_1 | on Permission Pairs
airflow-init_1 | sion Pairs to role Admin
airflow-init_1 | User "airflow" created with role "Admin"
airflow-init_1 | User "airflow" created with role "Admin"
airflow-init_1 | !!!!! Installing additional requirements: 'pymongo' !!!!!
airflow-init_1 | WARNING: This is a development/test feature only. NEVER use it in production!
airflow-init_1 | Instead, build a custom image as described in:
airflow-init_1 | https://airflow.apache.org/docs/docker-stack/build.html
airflow-init_1 | Adding requirements at container startup is fragile and is done every time
airflow-init_1 | the container starts, so it is only useful for testing and trying out
airflow-init_1 | of adding dependencies.
airflow-init_1 | Defaulting to user installation because normal site-packages is not writeable
airflow-init_1 | Collecting pymongo
airflow-init_1 | Downloading pymongo-4.10.1-cp38-cp38-manylinux_2_17_x86_64.manylinux2014_x86_64.whl.metadata
airflow-init_1 | (2.2 kB)
airflow-init_1 | Requirement already satisfied: dnspython<3.0.0,=>1.16.0 in /home/airflow/.local/lib/python3.8/
airflow-init_1 | site-packages (from pymongo) (2.4.2)
airflow-init_1 | Downloading pymongo-4.10.1-cp38-cp38-manylinux_2_17_x86_64.manylinux2014_x86_64.whl (930 kB)
airflow-init_1 | _____ 930.2/930.2 kB 13.9 MB/s eta 0:00:00
airflow-init_1 | Installing collected packages: pymongo
airflow-init_1 | Successfully installed pymongo-4.10.1
airflow-init_1 | [notice] A new release of pip is available: 23.3.2 -> 25.0.1
airflow-init_1 | [notice] To update, run: python -m pip install --upgrade pip
airflow-init_1 | /home/airflow/.local/lib/python3.8/site-packages/airflow/configuration.py:812 DeprecationWarni
airflow-init_1 | ng: The sqlalchemy_conn option in [core] has been moved to the sqlalchemy_conn option in [database] - the old setti
airflow-init_1 | ng has been used, but please update your config.
airflow-init_1 | /home/airflow/.local/lib/python3.8/site-packages/airflow/configuration.py:724 DeprecationWarni
airflow-init_1 | ng: The auth_backend option in [api] has been renamed to auth_backends - the old setting has been used, but please up
airflow-init_1 | date your config.
airflow-init_1 | /home/airflow/.local/lib/python3.8/site-packages/airflow/configuration.py:747 DeprecationWarni
airflow-init_1 | ng: The auth_backend option in [api] has been renamed to auth_backends - the old setting has been used, but please up
airflow-init_1 | date your config.

```

```

airflow-init_1 | /home/airflow/.local/lib/python3.8/site-packages/airflow/configuration.py:761 FutureWarning: T
airflow-init_1 | he auth_backends setting in [api] has had airflow.api.auth.backend.session added in the running config, which is need
airflow-init_1 | ed by the UI. Please update your config before Apache Airflow 3.0.
airflow-init_1 | /home/airflow/.local/lib/python3.8/site-packages/airflow/configuration.py:738 DeprecationWarni
airflow-init_1 | ng: The sqlalchemy_conn option in [core] has been moved to the sqlalchemy_conn option in [database] - the old setti
airflow-init_1 | ng has been used, but please update your config.
airflow-init_1 | /home/airflow/.local/lib/python3.8/site-packages/airflow/settings.py:194 DeprecationWarning: T
airflow-init_1 | he sqlalchemy_conn option in [core] has been moved to the sqlalchemy_conn option in [database] - the old setting ha
airflow-init_1 | s been used, but please update your config.
airflow-init_1 | /home/airflow/.local/lib/python3.8/site-packages/airflow/models/base.py:71 DeprecationWarning:
airflow-init_1 | The sqlalchemy_conn option in [core] has been moved to the sqlalchemy_conn option in [database] - the old setting
airflow-init_1 | has been used, but please update your config.
airflow-init_1 | 2.8.1
airflow-init_1 | airflow-init_1 exited with code 0

```

Launch the containers

```

ubuntu@ip-172-31-46-112:~/datascientest-data-engineering/airflow$ docker-compose up -d
Building with native build. Learn about native build in Compose here: https://docs.docker.com/go/compose-native-build/
airflow_redis_1 is up-to-date
airflow_postgres_1 is up-to-date
Creating airflow_airflow-worker_1 ... done
Starting airflow_airflow-init_1 ... done
Creating airflow_airflow-scheduler_1 ... done
Creating airflow_airflow-webserver_1 ... done
Creating airflow_flower_1 ... done

```

Verify

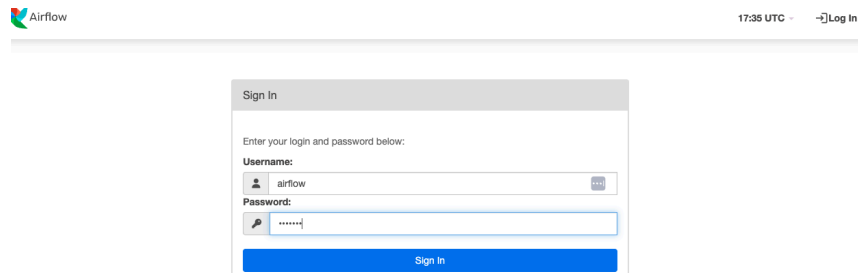
```

ubuntu@ip-172-31-46-112:~/datascientest-data-engineering/airflow$ docker container ls
CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS        PORTS
17a4eea1c93d   apache/airflow:2.8.1               "/usr/bin/dumb-init ..." 2 minutes ago   Up 2 minutes   0.0.0.0:55
55->5555/tcp, ::5555->5555/tcp, 8080/tcp   airflow_flower_1
66688557fdf0   apache/airflow:2.8.1               "/usr/bin/dumb-init ..." 2 minutes ago   Up 2 minutes   0.0.0.0:80
80->8080/tcp, ::8080->8080/tcp             airflow_airflow-webserver_1
a153df6dee96   apache/airflow:2.8.1               "/usr/bin/dumb-init ..." 2 minutes ago   Up 2 minutes   8080/tcp
d77c3bcf817c   apache/airflow:2.8.1               "/usr/bin/dumb-init ..." 2 minutes ago   Up 2 minutes   8080/tcp
b152939c3db4   apache/airflow:2.8.1               "/usr/bin/dumb-init ..." 10 minutes ago   Up 2 minutes   8080/tcp
e633a3744f2b   postgres:13                         "docker-entrypoint.s..." 11 minutes ago   Up 10 minutes   5432/tcp
51a998a694ac   redis:latest                       "docker-entrypoint.s..." 11 minutes ago   Up 10 minutes   0.0.0.0:63
79->6379/tcp, ::6379->6379/tcp             airflow_redis_1

```

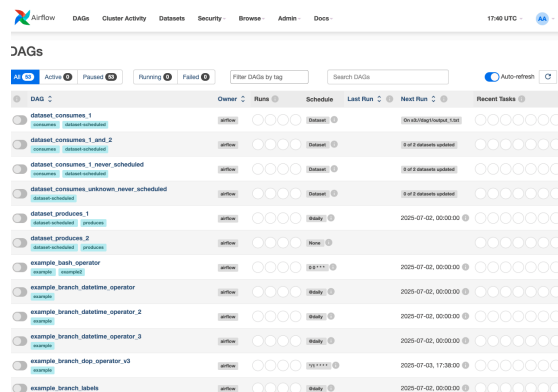
Open URL <http://108.129.115.158:8080/> and use default credentials:

- username airflow
- password airflow

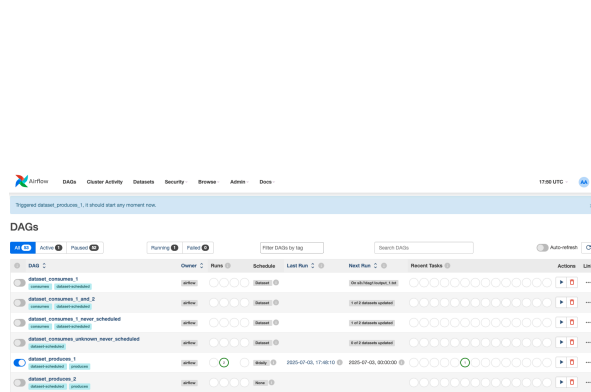


The image shows the Airflow Sign In page. At the top, it says "Sign In". Below that, it says "Enter your login and password below:". There are two input fields: "Username:" with the value "airflow" and "Password:" with masked characters "*****". At the bottom, there is a blue "Sign In" button.

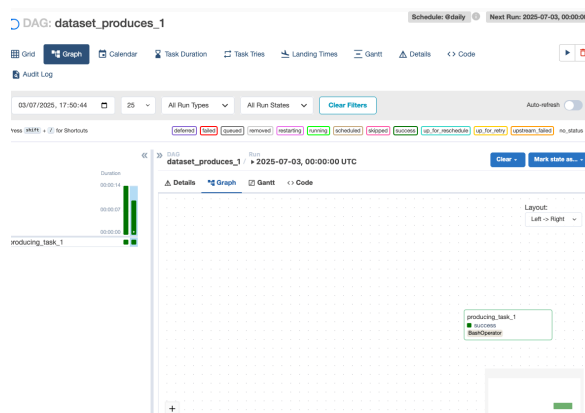
Exploring the Airflow user interface:



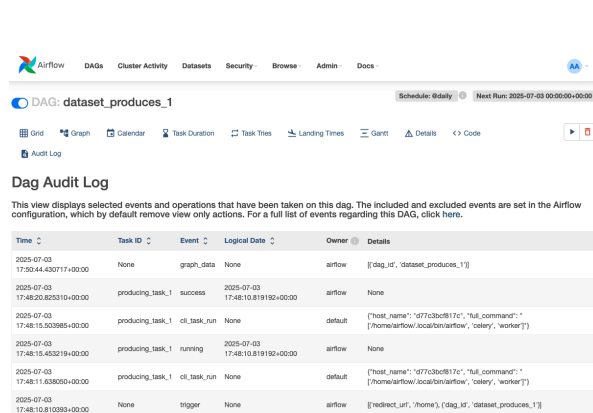
The image shows the Airflow DAGs list. It has a header with "Airflow" logo and navigation links: DAGs, Cluster Activity, Datasets, Security, Browse, Admin, Docs. The main section is titled "DAGs" and has a filter bar with "Active", "Paused", "Running", and "Failed" tabs. Below the filter bar, there is a table of DAGs. The table has columns: DAG, Owner, Runs, Schedule, Last Run, Next Run, and Recent Tasks. The DAGs listed include "dataset_consumes_1", "dataset_consumes_1_and_2", "dataset_consumes_1_never_scheduled", "dataset_consumes_unknown_never_scheduled", "dataset_produces_1", "dataset_produces_2", "example_branch_operator", "example_branch_datetime_operator", "example_branch_datetime_operator_2", "example_branch_datetime_operator_3", "example_branch_operator_v3", and "example_branch_labels".



This is a duplicate of the previous image, showing the Airflow DAGs list.



The image shows the Airflow DAG "dataset_produces_1". It has a header with "DAG: dataset_produces_1" and "Scheduler: @daily". Below the header, there is a navigation bar with "Grid", "Graph", "Calendar", "Task Duration", "Task Times", "Landing Times", "Gantt", "Details", and "Code". The "Graph" tab is selected. The graph shows a DAG with two tasks: "dataset_produces_1" and "producing_task_1". The "dataset_produces_1" task is a "graph_data" task and the "producing_task_1" task is a "cli_task_run" task. The graph shows the execution flow from "dataset_produces_1" to "producing_task_1".



The image shows the Airflow DAG Audit Log for "dataset_produces_1". It has a header with "DAG: dataset_produces_1" and "Scheduler: @daily". Below the header, there is a navigation bar with "Grid", "Graph", "Calendar", "Task Duration", "Task Times", "Landing Times", "Gantt", "Details", and "Code". The "Details" tab is selected. The details view shows a table of events. The table has columns: Time, Task ID, Event, Logical Date, Owner, and Details. The events listed are: "graph_data" (None), "producing_task_1" (success), "cli_task_run" (None), and "cli_task_run" (None).

The Airflow terminal:

Install and use Airflow CLI helper script

```
ubuntu@ip-172-31-46-112:~/datascientest-data-engineering/airflow$ wget https://dst-de.s3.eu-west-3.amazonaws.com/airflow_advance_fr/docker-compose/airflow.sh
--2025-07-03 17:54:55-- https://dst-de.s3.eu-west-3.amazonaws.com/airflow_advance_fr/docker-compose/airflow.sh
Resolving dst-de.s3.eu-west-3.amazonaws.com (dst-de.s3.eu-west-3.amazonaws.com)... 3.5.205.248, 3.5.204.58
Connecting to dst-de.s3.eu-west-3.amazonaws.com (dst-de.s3.eu-west-3.amazonaws.com)|3.5.205.248|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 1116 (1.1K) [binary/octet-stream]
Saving to: 'airflow.sh'

airflow.sh          100%[=====] 1.09K --.-KB/s in 0s

2025-07-03 17:54:55 (43.3 MB/s) - 'airflow.sh' saved [1116/1116]

ubuntu@ip-172-31-46-112:~/datascientest-data-engineering/airflow$ chmod +x airflow.sh
```

```
ubuntu@ip-172-31-46-112:~/datascientest-data-engineering/airflow$ ./airflow.sh bash
Creating airflow airflow-worker run ... done
```

[illegible]